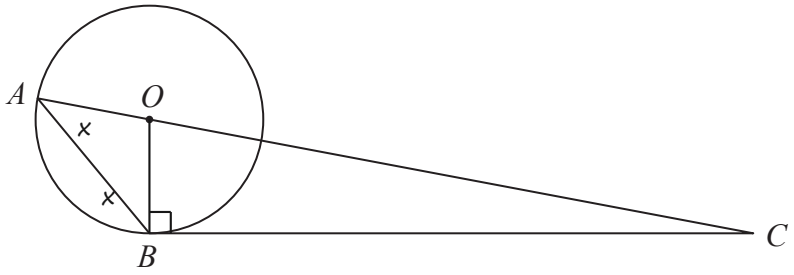
11



A and B are points on a circle, centre O.

BC is a tangent to the circle.
AOC is a straight line.
Angle $ABO = x°$.

Find the size of angle ACB, in terms of x.
Give your answer in its simplest form.
Give reasons for each stage of your working.

OA = OB, since both are radii.

So angle BAO = x°: base angles in an isosceles triangle are equal.

1 mark
Identify an unknown angle

Angle OBC = 90°: a tangent is perpendicular to the radius

at the point of contact.

1 mark
Link the tangent-radius reason

Angle ABC = (90 + x)°.

In triangle ABC, the angles add to 180°.

Angle ACB = 180° – x° – (90° + x°)

1 mark
Complete a method for the required angle

= (90 – 2x)°

1 mark
Simplified angle

1 mark
Complete linked reasons for the triangle method

(Total for Question 11 is 5 marks)